Ch2 – Hierarchical Clustering Technique

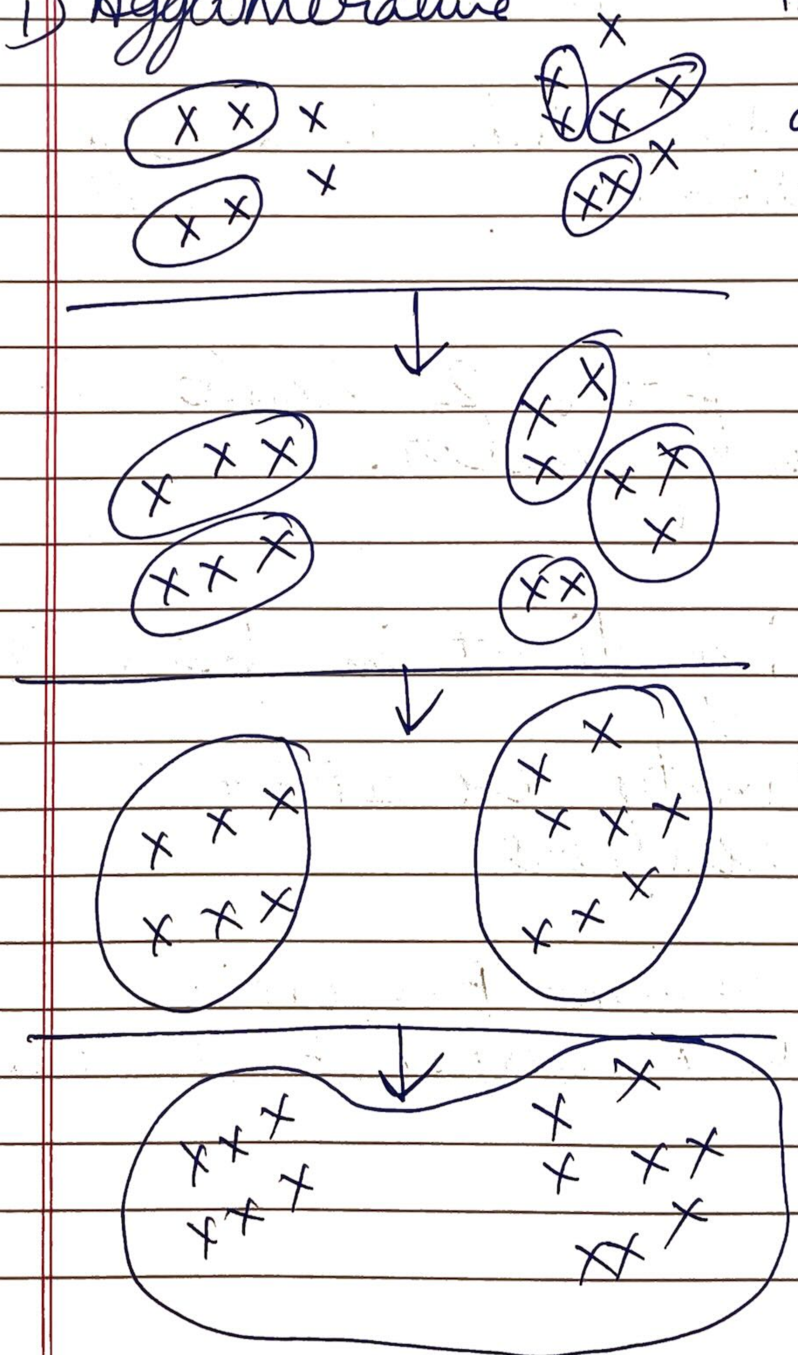
Video 1 : Agglomerative & Divisive Dendrograms

2 types of Hierarchical Clustering

1) Agglomerative
2) Divisive

1) Agglomerative

13 pts → 13 clusters
↓
9 clusters



5 clusters
↓
2 clusters
↓
1 cluster

Agglomerative → Grouping

Starts with individual pts and groups them until 1.

2) Divisive

It does the exact opposite from 1 cluster to 13 cluster.

How to divide is a big question.

Out of the 2 Agglo is more popular.

Key ingredient in both is Similarity or distance b/w clusters.

Dendrogram - Tree which shows merger/split of clusters

Video 2: Agglomerative Clustering

1. Compute the proximity matrix
2. Let each data point be a cluster
3. Repeat
4. Merge the two closest clusters
5. Update the proximity matrix
6. Until only a single cluster remains.

Similarity / Distance / Proximity b/w 2 clusters

	P1	P2 ∪ P3	P4	P5	....
P1	X	....	X	X	
P2 ∪ P3	....	....	....	....	....
P4	X	....	X	X	
P5	X	....	X	X	
⋮		....			

X – Already Calculated
.... – Recompute.

Date
Page 18

Video 3: Proximity methods : Advantages & Limitation

1 MIN

$$Sim(C_1, C_2) \downarrow$$

min $Sim(P_i, P_j)$ such that $P_i \in C_1$
$P_j \in C_2$

Take pairwise similarity

2 MAX

$$Sim(C_1, C_2) \downarrow$$

Max $Sim(P_i, P_j)$ such that $P_i \in C_1$
$P_j \in C_2$

3 GROUP AVERAGE

$$Sim(C_1, C_2) = \left[\sum_{\substack{P_i \in C_1 \\ P_j \in C_2}} Sim(P_i, P_j)\right] \div \left[|C_1| \times |C_2|\right]$$

Size of C_1 — Size of C_2

4 DISTANCE B/w CENTROIDS

MIN, MAX and GROUP AVG. can be trivially kernalized.

MIN NOTE: Smallest distance in proximity matrix, those points become one cluster.

MAX NOTE: Largest distance in proximity matrix, those points become one cluster.

GROUP AVERAGE

Imp points

1) Compromise b/w Single and Complete Link.
2) Strengths:-
Less susceptible to noise and outliers
3) Limitations:-
Biased towards globular clusters.

4 Ward's Method

1) Similarity of two clusters is based on the increase in squared error (SSE) when two clusters are merged
• Similarity to group average if distance between points is distance squared
2) Less susceptible to noise and outliers
3) Biased towards globular clusters
4) Hierarchical analogue of K-means
• Can be used to initialize K-means.

Video 4: Time and Space Complexity for Hierarchical Clustering

Space: $O(n^2) \rightarrow$ Similarity matrix
$n \rightarrow$ No. of data points
alot when n is large

Time: $O(n^3)$:- Atmost n iterations
Using Brute Force
In each iteration, updating similarity matrix ($O(n^2)$)
$\approx O(n^3)$

$O(n^2 \lg n)$ is possible but using different approach.

eg $\rightarrow$ For 10M points $\rightarrow 10^6$
$O(n^2) \Rightarrow 10^{12} = 1000$ Gb
≈ 1 Tb
in RAM is too much

So agglomerative is not very useful.

Video 5 **Limitations of Hierarchical clustering**

1) No mathematical functions. No optimization problem that we are directly solving.

2) Whether using MIN has problem with outliers, MAX breaks large clusters, can't accomodate different sized clusters and GROUP AVG has problems.

M.IMP 3) Space and Time Complexity.

Time $O(n^2 \lg n)$ OR $O(n^3)$

Space $O(n^2)$